Algebra: Partial Fractions & Division

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Decompose the rational expression $\frac{5x^2 - 3x + 2}{(x-1)(x^2+1)}$ into partial fractions.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

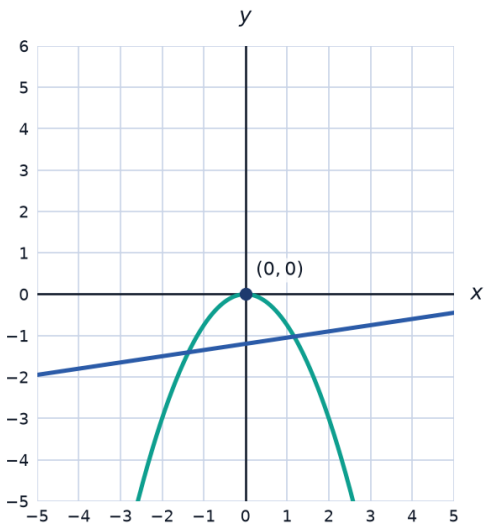
[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Algebra, Proof & Complex Numbers · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

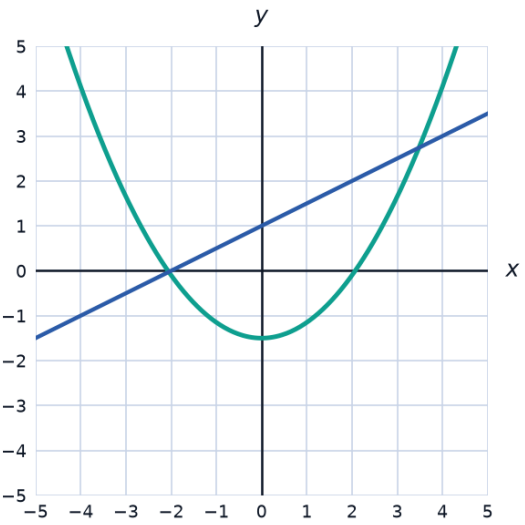


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

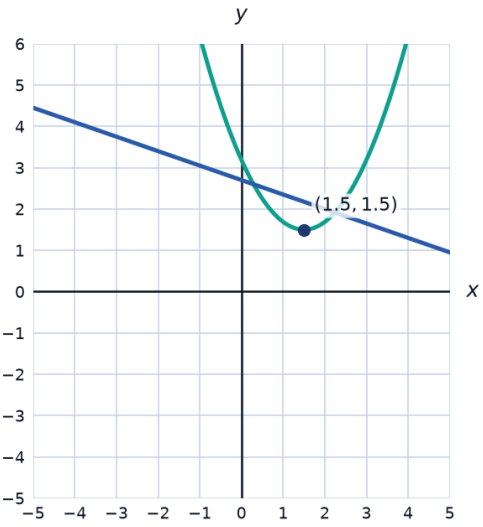
8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

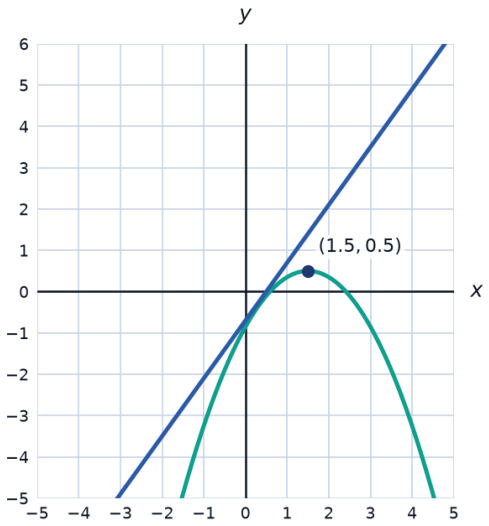
Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



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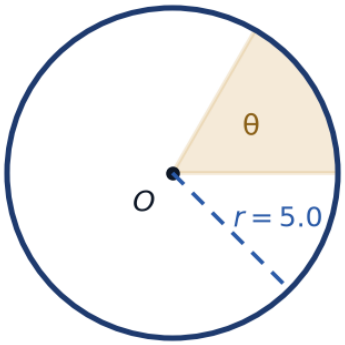
10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Complex Numbers

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Numerical Methods

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Differential Equations

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Poisson Distribution

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 13.01 Continuous Random Variables

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Divide $2x^3 - 3x^2 + 4x - 5$ by $(x - 2)$ to find the quotient and remainder.

a. Quotient $= 2x^2 + x + 6$, Remainder $= 7$

b. Quotient $= 2x^2 - 7x + 18$, Remainder $= -41$

c. Quotient $= 2x^2 + x + 6$, Remainder $= -7$

d. Quotient $= 2x^2 + 7x + 18$, Remainder $= 31$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express the improper rational function $\frac{x^2 + 5x + 7}{x + 2}$ in the form $Ax + B + \frac{C}{x + 2}$.

a. $x + 3 + \frac{1}{x + 2}$

b. $x + 3 + \frac{7}{x + 2}$

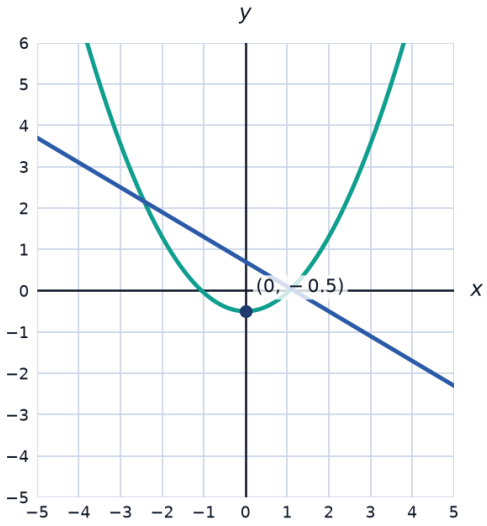
c. $x + 7 - \frac{7}{x + 2}$

d. $x + 2 + \frac{3}{x + 2}$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When $P(x) = x^4 + ax^2 + bx - 4$ is divided by $(x - 1)$, the remainder is 2; when divided by $(x + 2)$, the remainder is -8. Find a and b.



a. $a = 1, b = 4$

b. $a = 2, b = 3$

c. $a = -1, b = 6$

d. $a = 3, b = 2$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the correct partial fraction decomposition template for $\frac{3x^2 + 1}{(x - 2)^2(x + 3)}$?

a. $\frac{A}{x - 2} + \frac{B}{(x - 2)^2} + \frac{C}{x + 3}$

b. $\frac{A}{(x - 2)^2} + \frac{B}{x + 3}$

c. $\frac{Ax + B}{(x - 2)^2} + \frac{C}{x + 3}$

d. $\frac{A}{x - 2} + \frac{B}{x + 3}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the decomposition $\frac{4x}{(x - 1)^2} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2}$, find A and B.

a. $A = 4, B = 4$

b. $A = 4, B = 0$

c. $A = 0, B = 4$

d. $A = 2, B = 4$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the correct partial fraction template for $\frac{2x+5}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$

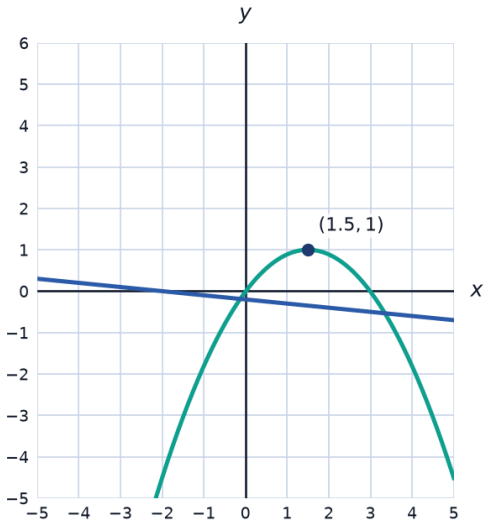
b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

c. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

d. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In $\frac{7x^2+5x+13}{(x-1)(x^2+4)}=\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$, find the value of A.



a. 5

b. 7

c. 2

d. 3

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For the decomposition in the previous question ($A=5$), find B and C.

a. $B=2, C=7$

b. $B=2, C=-7$

c. $B=7, C=2$

d. $B=-2, C=7$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

To expand $f(x) = \frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x for small $|x|$, which partial fraction form should be expanded?

a. $-\frac{2}{1-x} + \frac{5/2}{1+x/2}$

b. $\frac{2}{x-1} + \frac{5}{x+2}$

c. $\frac{2}{1-x} + \frac{5}{1+x/2}$

d. $-\frac{2}{1-x} + \frac{5}{1+x}$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the constant term in the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $\frac{1}{2}$

b. $-\frac{1}{2}$

c. -2

d. 5

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

State the radius of convergence for the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $|x| < 1$

b. $|x| < 2$

c. $|x| < 1/2$

d. $|x| < 7$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Partial fractions: if denominator has irreducible quadratic $x^2 + bx + c$, the numerator is:

a. A

b. $A/(x^2 + bx + c)$

c. $Ax^2 + B$

d. $Ax + B$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

$\int \frac{1}{x(x-1)}\,dx$ by partial fractions:

a. $\ln|x| - \ln|x-1| + C$

b. $\ln|x(x-1)| + C$

c. $\frac{1}{x-1/2} + C$

d. $\ln|x-1| - \ln|x| + C$

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

a. $(0.5, 0.5)$

b. $(0, 0)$

c. $(-1, -1)$

d. $(1.5, 1.5)$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Quadratic factor in partial fractions: $\frac{1}{x^2+4}$ gives integral:

a. $\frac{1}{2}\arctan(x/2) + C$

b. $\arctan x + C$

c. $\ln(x^2 + 4) + C$

d. $1/(2x + 4) + C$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

$\int \frac{1}{x^3 - x}\,dx$: partial fractions decomposition is:

a. $\frac{A}{x^2+1} + \frac{B}{x-1}$

b. $\frac{A}{x^3} + \frac{B}{x}$

c. $\frac{Ax+B}{x^2} + \frac{C}{x-1}$

d. $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Sum of series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$ (partial fractions $\frac{1}{2}(\frac{1}{n-1} - \frac{1}{n+1})$):

a. 1

b. 3/4

c. 3/2

d. 5/4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Differential Equations

Partial fractions for $\frac{1}{(x+1)^2}$ uses:

a. repeated linear factor

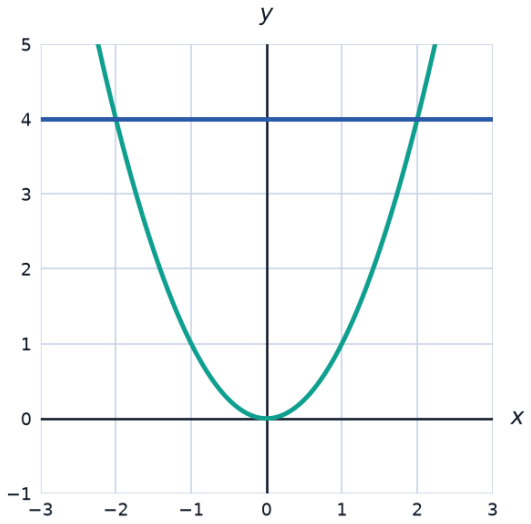
b. linear factors

c. irreducible quadratic

d. none

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Poisson Distribution

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?



a. $\frac{32}{3}$

b. $\frac{16}{3}$

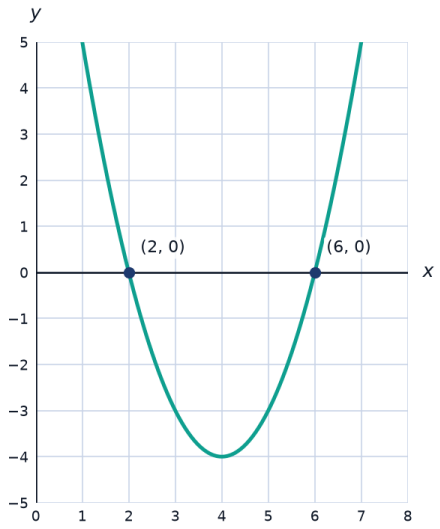
c. 8

d. 16

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 13.01 Continuous Random Variables

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.



a. 12

b. 16

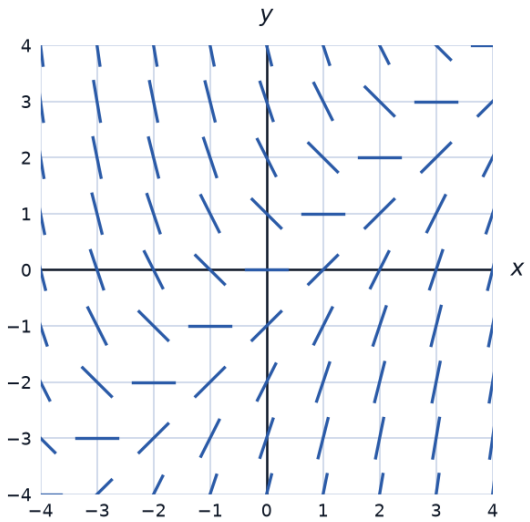
c. 20

d. 24

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 14.01 Sampling & Estimation

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$



a. $\frac{dy}{dx} = \frac{x}{y}$

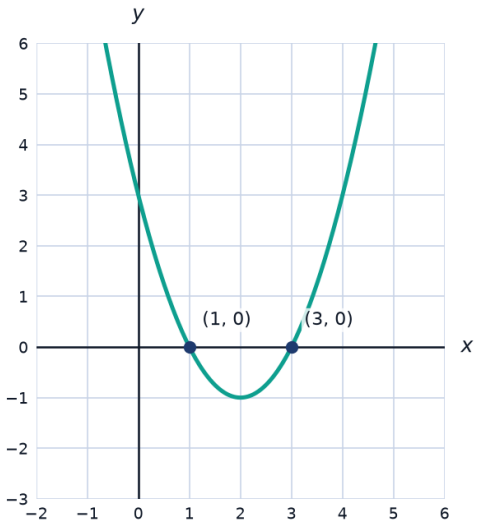
b. $\frac{dy}{dx} = \frac{y}{x}$

c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 15.01 Hypothesis Testing

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.



a. 4

b. 3

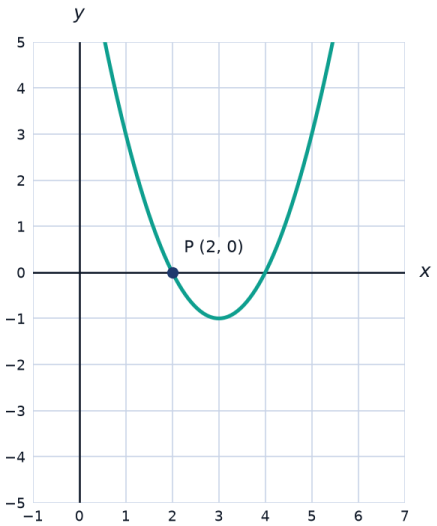
c. 1

d. -3

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Exponentials & Logarithms

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.



a. 1

b. 2

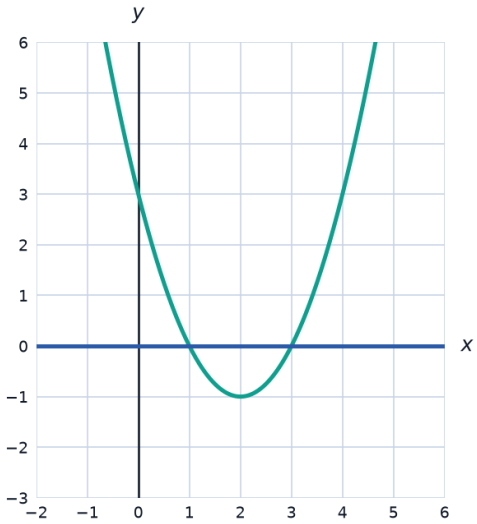
c. 4

d. 8

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Trigonometry

Consider the quadratic function $f(x) = x^2 - 4x + 3$. Find the value of x where the graph of $f(x)$ intersects the x-axis at the point $(x, 0)$, given that $x > 1$.



a. 2

b. 3

c. 4

d. 5

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Differentiation

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$

c. 16π

d. $\frac{64\pi}{3}$

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